

Demonstration of Full-network Level Digital Twin for QoT Estimation in Field-deployed ROADM Networks

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Abstract We propose and demonstrate a full-network level digital twin framework for QoT estimation in China Telecom's field-deployed ROADM networks, achieving mean absolute errors of 0.53 dBm power and 0.68 dB OSNR for 95% test samples across all frequencies and loading rates. ©2026 The Author(s)

Introduction

Driven by the surge of large-scale AI applications and rapid expansion of data centers, the operation and maintenance (O&M) of optical networks are evolving towards higher-level digitalization and automation. As the key element in realizing digital twin (DT) of optical networks, accurate estimation of the quality of transmission (QoT) is pivotal for promoting high-level intelligent applications in O&M [1]. However, current QoT estimation methods are mainly proposed for system-level scenarios, including end-to-end links [2] and small-scale networks [3]. The applicability of various methods remains limited for QoT estimation in the large-scale or even full-network level optical networks. Furthermore, field-deployed networks suffer from non-ideal factors such as complex environmental impacts, device performance variations, and unknown insertion losses, which are negligible or measurable in simulations and laboratory scenarios, making network-level QoT estimation more challenging in field-deployed networks.

To achieve network-level DT of optical networks, analytical models, such as Gaussian noise (GN) model [4] and enhanced GN (EGN) models [5], have demonstrated their accuracy of QoT estimation in laboratory [6] and small-scale field-deployed networks [7]. However, laborious parameter refinements are usually required to accurately characterize the physical processes within each device, which is not feasible for large-scale field-deployed networks with a massive number of parameters and significant device differences. Machine learning-based methods have achieved high accuracy in modelling individual devices [8] or optical multiplex sections (OMSs) [9], but the modelling errors often accumulate due to cascading and unmeasurable insertion loss in chained networks. Recently, parameter refinement [10] and cascade learning [11] have been proposed to reduce the cascading errors, but they have not yet been validated in

large-scale networks. Moreover, large-parameter neural network [12] and variable-length transformer [13] have been applied to QoT estimation in large-scale networks, but they do not consider non-ideal factors in practical. Therefore, high-fidelity DT methods for QoT estimation in large-scale field-deployed networks is still missing.

In this paper, a full-network level DT framework for QoT estimation in field-deployed ROADM networks is proposed and demonstrated in China Telecom's two field-deployed ROADM networks. By constructing general DT models for transmission simulation and impairments calculation in OMSs and ROADM nodes, accurate QoT estimation is achieved along any optical channel (OCh) path. Field validation results show that 0.53 dBm power and 0.68 dB OSNR mean absolute errors (MAEs) are achieved for 95% test samples. Further analysis illustrates that the DT framework shows accurate and stable performance across all frequencies and loading rates, exhibiting great potential in O&M of optical networks.

Full-network Level DT for QoT Estimation

For any service transmitted in the ROADM networks, a general DT framework is constructed for accurate QoT estimation along its OCh path, as shown in Fig.1(a). In this framework, the power spectrum and OSNR of OMSs and impairments of ROADM nodes are respectively modelled by DT models for OMS and ROADM, making the entire transmission process along the OCh path visible and transparent.

For each OMS, the evolutions of its power spectrum and OSNR primarily depend on the configurations of each optical line amplifier (OLA) node it contains, including gain G , attenuation of variable optical attenuator L , input power P_{in} and output power P_{out} . These configurations are merged into a one-dimensional configuration

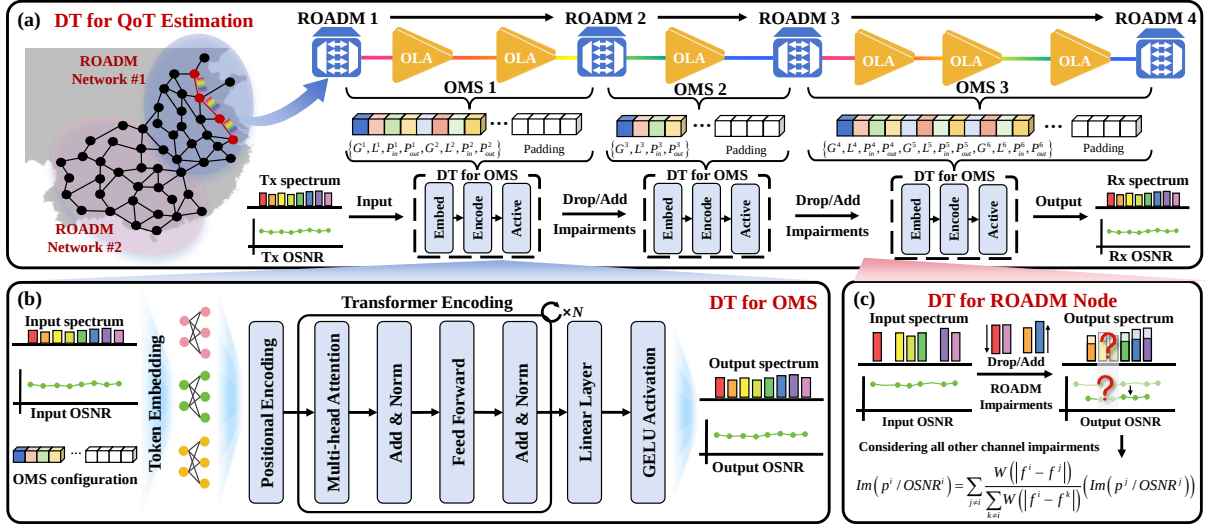


Fig. 1: (a) Schematic of full-network level DT framework for QoT estimation in China Telecom's two field-deployed ROADM networks. (b) DT model for power spectrum and OSNR estimation in each OMS. (c) DT model for drop/add services and channel impairments calculation in each ROADM Node.

vector, which serves as the input to the DT model for OMS, as shown in Fig. 1(a). Since different OMSs contain different numbers of OLA nodes, we consider adding padding at the end of configuration vectors to make their lengths uniform. In the DT model for OMS shown in Fig. 1(b), the input spectrum, OSNR, and OMS configuration are treated as independent tokens for embedding. The embedded tokens are then positionally encoded and input to N cascaded Transformer encoder blocks. Each block consist of a multi-head attention layer, a feed forward layer, and two addition and normalization layers, fully capturing correlations between QoT and configuration. The output of encoder blocks is fed to a linear layer and activated by GELU function to obtain output spectrum and OSNR of the OMS.

When transmission passes through ROADM nodes, unknown insertion loss and wavelength-dependent impairments often occur due to the inherent imperfections and differences of devices, including wavelength selective switches, filters, and connectors. To estimate these impairments, a DT model considering both drop/add services and ROADM impairments is constructed, as shown in Fig. 1(c). The impairments of different channels are separately calculated in each ROADM. Specifically, for the i -th channel, its power/OSNR impairment $Im(p^i/OSNR^i)$ can be estimated through multiplying the impairments of other channels by a weight function $W(\cdot)$, where the impairment of each channel is normalized. Here, $W(\cdot)$ should be a monotonically decreasing function of the frequency difference $\Delta f = |f^i - f^j|$, which assigns higher weights to impairments of adjacent channels.

Field-trial Setup

To verify the feasibility and effectiveness of the

proposed DT framework for full-network level QoT estimation in field-deployed networks, we study China Telecom's two field-deployed networks shown in Fig. 1(a), covering over 200 ROADM nodes and 700 OLA nodes in 19 provinces and municipalities. The two ROADM networks are bi-directionally constructed, fully covering the 96 channels in C-band (191.3–196.1 THz) with 50 GHz channel spacing. Each service is modulated using polarization division multiplexing-quadrature phase shift keying (PDM-QPSK) format at a symbol rate of 28 Gbaud. The optical channel monitors are deployed at the optical preamplifiers and optical booster amplifiers of each ROADM node to monitor power and OSNR of each channel, and configurations of each OLA nodes (G, L, P_{in}, P_{out}) are obtained from the network controller. Over 2000 OCh-level service information are collected, including configurations, power spectra, and OSNR of each OMS and ROADM node along the entire OCh path.

Results and Discussions

In the DT model for OMS, the input power spectrum, OSNR, as well as OMS configuration are embedded to 256 dimensions and fed to $N = 6$ Transformer encoder blocks. In each block, the multi-head attention layer contains 8 attention heads, and the dimension of feed forward layer is 1024. Moreover, the Adam optimizer is adopted for training 1000 iterations, with the learning rate initially set to 10^{-4} and decreased to 90% of the previous value every 100 iterations. Moreover, the weight function $W(\cdot)$ is set to a Gaussian function (i.e., $W(\Delta f) = \exp(-\Delta f^2 / 2\sigma^2)$) in the DT model for ROADM nodes, which changes smoothly when Δf is small. σ is determined through optimization on the training set.

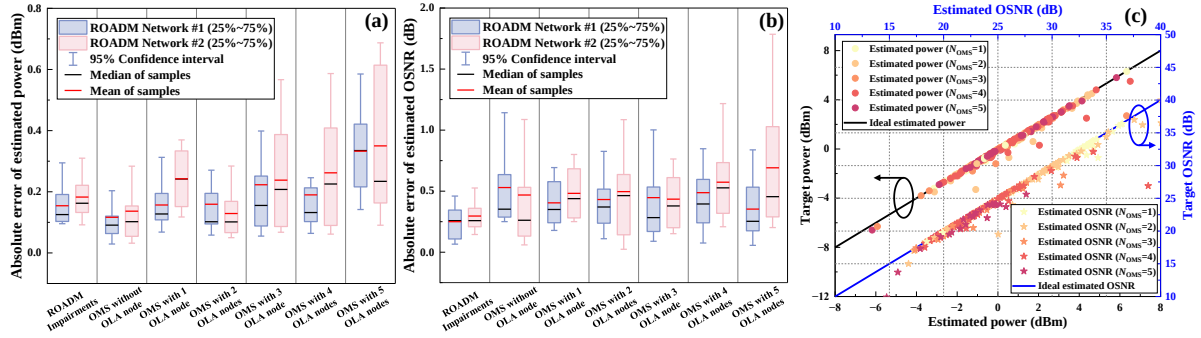


Fig. 2: (a) Power and (b) OSNR estimation results of DT models for ROADM nodes and OMS. (c) OCh-level power and OSNR estimation results of the DT framework in China Telecom's two ROADM networks.

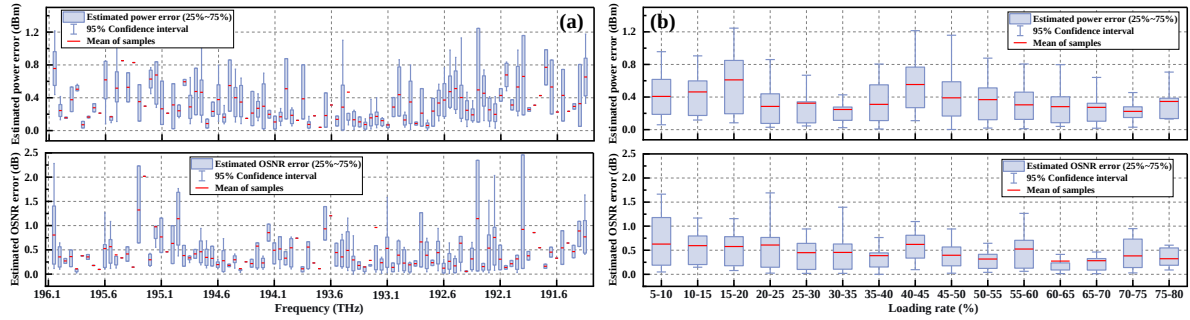


Fig. 3: OCh-level power and OSNR estimation results at different (a) frequencies and (b) average loading rates.

After sufficient training and optimization, the predicted power and OSNR results of the DT models for OMS and ROADM on the test sets of ROADM Network #1 and ROADM Network #2 are shown in Fig. 2(a) and Fig. 2(b). Since ROADM network #2 contains more ROADM nodes and OMSs, it faces more complex factors during transmission, thus exhibiting higher power and OSNR errors in most scenarios. Moreover, OMSs with more than 2 OLA nodes account for only about 20% of the dataset, and the DT model also exhibits larger 95% confidence intervals and MAEs in these OMSs. Nevertheless, the median power and OSNR errors in both networks do not exceed 0.32 dBm and 0.52 dB, respectively.

Based on these established DT models, the OCh-level estimation results of two networks are further verified, as shown in Fig. 2(c). It is observed that the proposed DT framework can accurately estimate the channel power ranges from -6.6 dBm to 6.3 dBm and the OSNR ranges from 10 dB to 40 dB of OCh-level services with different number of OMSs N_{OMS} . When estimating QoT of services with large or small target power or OSNR, it tends to exhibit larger error, as these extreme samples are rarely seen in the dataset. Only about 3% of all test samples show large power errors over 1.5 dBm and OSNR errors over 2.5 dB, where most of which are samples containing 3 or more OMSs suffering from more cascading errors. Nevertheless, MAEs of power and OSNR for test samples in 95% confidence interval are 0.53 dBm and 0.68 dB, respectively.

Furthermore, we investigate the performance of the DT framework at different frequencies and average loading rates along the OCh path to analyse its versatility in different scenarios, as shown in Fig. 3(a) and Fig. 3(b). Each frequency included at least one test sample to ensure the all frequencies are covered. In all 96 frequencies, the channels exhibited similar estimation errors, especially in most channels in the range of 192.6–194.6 THz. Larger errors are more likely to occur in channels with few samples and edge channels more susceptible to effects of non-ideal devices. For different loading rates, the proposed DT framework demonstrated stable power and OSNR estimation results under various loading conditions. Particularly, the estimation error is lower in scenarios with higher loading rate, where ROADM impairments calculated from more channels are more accurate.

Conclusions

A full-network level DT framework was proposed and demonstrated for QoT estimation in China Telecom's two field-deployed ROADM networks. By constructing DT models for OMS and ROADM nodes, accurate OCh-level QoT estimation was achieved, with 0.53 dBm and 0.68 dB MAEs of power and OSNR for 95% test samples, respectively. Further verifications at different frequencies and loading rates illustrate that the proposed DT framework can achieve accurate and stable QoT estimation in various scenarios, thereby empowering O&M of optical networks.

Acknowledgements

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